

Thermochemical processing for the sustainable disposal of spent filter waste

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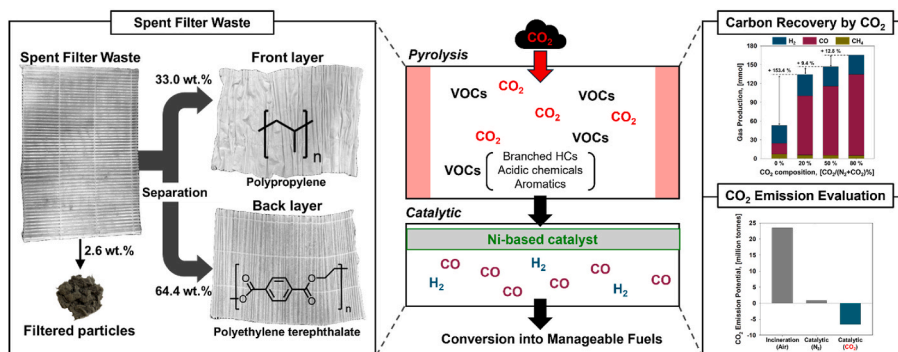
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HIGHLIGHTS

- Particles and constituent polymers in spent filter waste (SFW) were characterized.
- Employing CO₂ to pyrolysis identified its interaction with SFW-derived volatiles.
- Functionality of CO₂ for producing CO become stronger by adopting a Ni catalyst.
- Potential for the CO₂ emission of CO₂-mediated catalytic pyrolysis was evaluated.

GRAPHICAL ABSTRACT



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ABSTRACT

High-efficiency particulate air filters are widely used for indoor air purification. Spent filter waste (SFW), which can trap infectious and toxic substances, is primarily treated via incineration. This method causes environmental concerns, particularly regarding the generation of carbon dioxide (CO₂) and other air pollutants. To cope with these issues, this work proposes the pyrolysis of SFW using CO₂ as a sustainable alternative to conventional incineration. The introduction of CO₂ enhanced reactivity during pyrolysis, potentially offering a more sustainable process. The SFW consisted of filtered particles and two distinct filter/support layers, and the presence of toxic chemicals and primary polymer constituents was characterized. While CO₂ had a minimal impact on enhancing syngas production, owing to its slow reaction rate during SFW pyrolysis, adding a nickel-based catalyst significantly improved CO₂ reactivity. This resulted in a 649.7% increase in carbon monoxide (CO) production compared to that in pyrolysis under N₂. The potential of this pyrolysis system for reducing CO₂ emissions was evaluated against that of conventional incineration. Overall, this study presents a promising method for the pyrolytic conversion of SFW into combustible gases, particularly CO, while leveraging CO₂ utilisation to mitigate global warming.

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1. Introduction

Respiratory diseases, including Coronavirus Disease and Chronic Obstructive Pulmonary Disease, are major contributors to global mortality (WHO, 2024b). In 2021, these diseases were responsible for the deaths of 15.5 million people, accounting for 22.7% of the total global mortality (WHO, 2024a). They are caused by the inhalation of air contaminated with airborne pathogens, including viruses and bacteria, and particulate matters (Wang et al., 2021). An increase in the population density has further facilitated their spread (Freeman et al., 2020). To address this issue, air filtration systems have been implemented in indoor environments such as households, offices, and industrial facilities (Lowther et al., 2020). Air filters are typically categorised into four grades based on their particle filtration efficiency: pre-filters, medium-efficiency filters, high-efficiency particulate air (HEPA) filters, and ultra-low-penetration air filters (Liu et al., 2017). The use of HEPA filters is particularly aimed at capturing pathogens (Atalie et al., 2024), enabling to remove 99.97% of particles with diameters $\geq 0.3 \mu\text{m}$ (Montaluisa-Mantilla et al., 2023). In 2021, the global market value for HEPA filters was USD 648 million, with projections indicating a doubling of the market within the next decade (Insights, 2024). Consequently, the generation of spent filter waste (SFW) is expected to increase. Given that SFW can harbour infectious and toxic substances that pose considerable health risks, finding viable methods to safely degrade or dispose of this waste is crucial. The biodegradation of SFW is limited by its recalcitrant nature, as it is typically composed of glassy and polymeric fibres (Madireddi et al., 2021).

Thermochemical processing may be a strategic method for SFW disposal. Indeed, incineration is the predominant practice for rapidly reducing the volume of carbonaceous materials while recovering energy (Luo et al., 2024). This process effectively destroys filtered hazardous substances through thermal decomposition and generates substantial energy from the combustion of polymeric fabrics, similar to that of conventional transportation fuels such as gasoline and diesel (Vlasopoulos et al., 2023). However, incineration presents environmental challenges, because the large molecular structure of the polymeric components complicates the optimisation of the air-to-fuel equivalence ratio (Chughtai et al., 2002). Incomplete combustion can lead to the release of hazardous air pollutants, including carbon monoxide (CO), volatile organic compounds (VOCs), and polycyclic aromatic hydrocarbons (Yang et al., 2023). Given these concerns, pyrolysis may serve as an alternative for the thermal decomposition of infectious and toxic substances in the absence of oxygen. This process also converts the polymeric fabrics in SFW into more manageable fuels, such as liquid and gaseous pyrolysis products (Gao et al., 2022). The char yield is typically negligible for thermoplastics (Sorolla-Rosario et al., 2022). The resulting pyrolytic products, particularly the gaseous fraction, offer more precise control over the equivalence ratio during combustion, promoting complete oxidation (Chughtai et al., 2002). However, the complex nature of the liquid product, which contains long-chain hydrocarbons and functionalised chemicals, poses challenges to its use in internal combustion engines (Ansari et al., 2021; Kusenberget al., 2022). Moreover, the need for extensive refinement to obtain liquid fuels with appropriate carbon chain lengths and chemical structures diminishes the economic feasibility of this approach (Yáñez et al., 2020).

Based on these considerations, transforming SFW into gaseous pyrolytic products is a more favourable strategy. Gasification has been a primary approach to produce syngas, a mixture of H_2 and CO, with recent research exploring advanced techniques incorporating microwave or plasma. However, the necessity of high gasifying temperature presents a significant obstacle to the widespread adoption of gasification. Therefore, a growing attention has been made for syngas enhancement in the pyrolysis temperatures of 300–700 °C with an assistance of reactive gas agents, particularly carbon dioxide (CO_2). Despite the inherent gasifying properties of CO_2 , its activation typically requires temperatures of ≥ 700 °C (Khosravi et al., 2023). The use of CO_2

in the pyrolysis process reveals a potential to serve as a partial oxidant. However, the mechanistic role of CO_2 remains partially understood. Thus, the main focus of this work was to elucidate the CO_2 's reactive functionality in the scope of pyrolysis test for SFW while assessing its effect on the composition of pyrolysis products, particularly to enhance syngas production. To promote CO_2 participation in the reaction, the conventional test setup was modified for catalytic pyrolysis, incorporating a heat block and two different catalysts: Al_2O_3 and $\text{Ni}/\text{Al}_2\text{O}_3$. Nickel (Ni) was selected because of the cost-effectiveness and strong ability to dissociate chemical bonds (Si et al., 2023). SFW pyrolysis experiments were conducted under 2 atm: N_2 and CO_2 . Resulting pyrolytic products generated under these conditions were qualitatively and quantitatively scrutinised. The comparative results provide insights into the reactive functionality of CO_2 , offering a novel strategy for enhancing carbon utilisation in pyrolysis systems.

2. Experimental

2.1. Chemical reagents

The chemicals used in this study, including dichloromethane (CH_2Cl_2 , 99.8%, Prod. No. 3030-2304), methanol (CH_3OH , 98.0%, Prod. No. 5558-3310), 1-octanol ($\text{C}_8\text{H}_{17}\text{OH}$, 98.0%, Prod. No. 5634-4405), nitric acid (HNO_3 , 70.0 %, Prod. No. 5620-4100), nickel (II) nitrate hexahydrate ($\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$, 97.0%, Prod. No. 5608-4405), and activated alumina (Al_2O_3 , Prod. No. 1038–1405) were acquired from Daejung Chemicals Co. (Korea). UHP-grade N_2 and CO_2 , as well as zero-grade air, were acquired from GreenGas Co. (Korea). A reducing gas comprising 20-vol.% H_2 and 80-vol.% N_2 , and a gas calibration mixture, comprising 20-vol.% H_2 , CO, CH_4 , CO_2 , and N_2 , were purchased from RIGAS Co. (Korea).

2.2. SFW preparation and characterisation

After the use of HEPA grade filter, SFW was sourced from the air purifier used in our laboratory. The filter material was cut into small pieces measuring $< 5 \times 5 \text{ mm}$. Before characterisation, filtered particles were removed from the SFW and then the pretreated SFW was separated into two distinct layers, such as front layer and back layer, via immersion in a CH_2Cl_2 solvent. The surface morphologies and elemental distributions of the layers were visualized using a scanning electron microscope unit (Cube-II Mini-SEM, EmCrafts, Korea) equipped with an energy-dispersive X-ray spectroscope (AZtecOne, Oxford Instruments, UK). The functional groups in each layer were determined using a Fourier transform-infrared spectrometer (FT-IR, Cary 630, Agilent, USA). Moreover, a proximate analysis of the SFW was carried out using a thermogravimetric analysis instrument (TGA, STA 449 F5 Jupiter, Netzsch, Germany), resulting in 0.29 wt% of moisture, 91.10 wt% of volatile matters, and 8.61 wt% of fixed carbon. For ultimate analysis of the HEPA filter waste, the elemental composition was measured using an elemental analyser (EuroVector, EA3000, Italy), confirming 71.03 wt% of carbon, 7.63 wt% of hydrogen, and 21.32 wt% of oxygen. The masses of the two separate layers were gravimetrically determined to be 33.9 wt % and 66.1 wt%, respectively.

To confirm the chemical constituents evolved from the thermolyses of the filtered particles, front layer, back layer, and SFW, pyrolysis-gas chromatography/mass spectrometry (Py-GC/MS) tests were performed. The Py-GC/MS system consisted of a micro-scale pyrolysis device (Rx-3050TR, Frontier Lab, Japan) and a GC/MS apparatus (8890/5977B, Agilent, USA). A sample ($100 \pm 0.1 \mu\text{g}$) was placed on a stainless cup and aligned with the pyrolysis device. The operating conditions are described in detail in a previous study (Huang et al., 2020). TGA of the SFW and two layers was performed using a TGA unit (STA 449 F5 Jupiter, Netzsch Co., Germany). A $10.00 \pm 0.01 \text{ mg}$ of sample was placed on Al_2O_3 crucible positioned in the middle of a reaction chamber. The temperature program for all TGA tests were from 35 to 900 °C at

steady rate of $10\text{ }^{\circ}\text{C min}^{-1}$. For each TGA run, the flow rate of N_2 , CO_2 , or air was 80 mL min^{-1} , simultaneously providing $20\text{ mL N}_2\text{ min}^{-1}$ as a balanced protective gas.

2.3. Catalyst synthesis and characterisation

A Ni-based catalyst was synthesized following a preparation method for an egg-shell-type catalyst (Valizadeh et al., 2021). For preparation of a Ni solution, dissolution of $\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ in deionised water was made. Prior to the precursor solution's impregnation, Al_2O_3 was heated at $300\text{ }^{\circ}\text{C}$ with isothermal time of 1 h to remove impurities, including moisture and organics. The pretreated Al_2O_3 was immersed with a $\text{C}_8\text{H}_{17}\text{OH}$, a hydrophobic reagent, for 24 h. After separating the saturated Al_2O_3 , its impregnation was performed using the Ni precursor solution while being maintained at $140\text{ }^{\circ}\text{C}$ for fast evaporation, resulting in surficial Ni fixation. To remove the remaining $\text{C}_8\text{H}_{17}\text{OH}$ from the impregnated sample, it was heated at $200\text{ }^{\circ}\text{C}$ for 3 h, noting that its boiling point was $195\text{ }^{\circ}\text{C}$ (Bringué et al., 2017). Calcination of the prepared catalyst was performed under air conditions and using the following temperature programs from ambient ($25\text{ }^{\circ}\text{C}$) to $800\text{ }^{\circ}\text{C}$ at $2\text{ }^{\circ}\text{C min}^{-1}$ and $800\text{ }^{\circ}\text{C}$ for 3 h. Subsequently, reduction of the calcined one was conducted under 20 vol% H_2 (N_2 balance) using the identical temperature programs as with calcination. To confirm the Ni loading in $\text{Ni}/\text{Al}_2\text{O}_3$, it was acid-extracted using a microwave digester (ETHOS 1600, Milestone, USA). The extractive solution was tested by an inductively coupled plasma-optical emission spectroscopy unit (ICP-OES, 5800, Agilent, USA), determining a 4.94 wt% of Ni.

After the catalytic pyrolysis from N_2 and CO_2 conditions, two used catalysts were retrieved. The temperature-programmed oxidation (TPO) tests for the pristine catalyst and two used $\text{Ni}/\text{Al}_2\text{O}_3$ were performed using a TGA unit. A $50.00 \pm 0.02\text{ mg}$ of catalyst was loaded on a sample carrier, and the reactor was purged with $80\text{ mL air min}^{-1}$ and $20\text{ mL N}_2\text{ min}^{-1}$. The TGA instrument was operated at two consecutive steps: (1) impurities removal (from 35 to $200\text{ }^{\circ}\text{C}$ with $20\text{ }^{\circ}\text{C min}^{-1}$ and then $200\text{ }^{\circ}\text{C}$ for 30 min) and (2) TPO test (from 200 to $900\text{ }^{\circ}\text{C}$ with $20\text{ }^{\circ}\text{C min}^{-1}$). During the TPO test, oxidation of deposited carbon on the used catalysts led to the mass reduction.

2.4. Pyrolysis test setups

For pyrolysis of the SFW, two pyrolysis test setups, conventional and catalytic, were used. The conventional test setup was equipped with a cylindrical quartz reactor (1000-mm length and 25.4-mm outer diameter with 3-mm thickness) and a tubular furnace (with two heat zones). The temperatures of the two heat zones were adjusted independently. In conventional pyrolysis, a heat zone of a tubular furnace was heated from 100 to $700\text{ }^{\circ}\text{C}$ at $10\text{ }^{\circ}\text{C min}^{-1}$. In addition, a $3.00 \pm 0.01\text{ g}$ of sample was loaded in the middle of first heat zone. To construct catalytic pyrolysis, the first heat zone was operated identically to the conventional setup. The second heat zone was maintained at $700\text{ }^{\circ}\text{C}$. An Al_2O_3 or $\text{Ni}/\text{Al}_2\text{O}_3$ catalyst with $3.00 \pm 0.03\text{ g}$ was placed in the middle of second heat zone. All the pyrolysis runs were carried out in 2 atm: N_2 and CO_2 . The totally purged flow rate for both conditions was set as 800 mL min^{-1} . The CO_2 conditions were made by providing a reference N_2 gas (160 mL min^{-1}), equivalent to 80-vol.% CO_2 . The addition of 20-vol.% N_2 under the CO_2 conditions was intended to serve as an internal standard gas due to its inertness during the pyrolysis test.

2.5. Pyrolytic product analysis

The pyrolysis of SFW produced VOCs containing condensable (pyrolytic oil) and non-condensable (pyrolytic gas) products. Effluents from the pyrolysis reactor were passed through a condensation system, maintained at $-40\text{ }^{\circ}\text{C}$, to collect the pyrolytic oil. The condensed oil was dissolved in CH_2Cl_2 (100 mL) and CH_3OH (100 mL), and the extracted solution was analysed using a GC/MS unit equipped with a DB-5MS

capillary column. Non-condensable pyrolytic gases were directly injected into a micro-gas chromatograph (micro-GC) (990, Agilent, USA) equipped with a Thermal Conductivity Detector. The micro-GC was constructed with Molseive-5A and a PLOT-U columns. To quantify the key pyrolytic gases, including H_2 , CO , CH_4 , CO_2 , and N_2 , a gas mixture standard with 20-vol.% for each gas was employed.

3. Results and discussion

3.1. Characterisations of SFW

Commercial HEPA filters, designed to remove airborne particles sized $\geq 0.3\text{ }\mu\text{m}$, typically consist of multiple layers of various polymeric fabrics (Sheraz et al., 2023). Each polymer layer contributes to particle capture owing to its porous structure and distinct pore morphology (Yoon et al., 2012). The filtered particles were removed from the SFW, representing 2.6 wt% of the total mass. The pretreated filter was dissolved in a CH_2Cl_2 solution to isolate the fabric components of the SFW, revealing two distinct front and back layers. Gravimetric analysis revealed that the front layer constituted 33.0 wt% of the SFW mass, while the back layer accounted for 64.4 wt% (Fig. 1(a)).

To examine the structure of the two distinct layers, the surface morphology and elemental distribution were analysed using SEM/EDX (Fig. 1(b)). The front layer exhibited randomly arranged fibres, forming densely packed pores ranging from 0.5 to $2.0\text{ }\mu\text{m}$. This structure suggested that the front layer functioned as the primary filter for capturing the targeted particles. By contrast, the back layer displayed a more organised fibre arrangement with larger pores, indicating its role in structural support. Elemental analysis of the two layers (Fig. 1(b)) revealed that the front layer is composed solely of carbon, whereas carbon and oxygen were present in the back layer. This suggested that the front layer might have had a hydrocarbon-based polymer backbone, whereas the back layer likely contained oxygen-containing functional groups.

FT-IR analysis was conducted to identify the exact polymer types in the filter and support layers of the SFW. FT-IR is a highly effective analytical tool for detecting functional groups associated with monomeric molecules. Fig. 1(c) shows the FT-IR spectra of the filter and support. The FT-IR analyses of polypropylene (PP) and polyethylene terephthalate (PET) are also incorporated as references in Fig. 1(c). The FT-IR spectrum for the filter developed five noticeable peaks, at 2955 , 2924 , 2842 , 1459 , and 1379 cm^{-1} (Tariq et al., 2020), closely matching the PP reference. This observation suggested that the filter was composed of a PP-based fabric. Similarly, six characteristic peaks (at 1739 , 1267 , 1120 , 1046 , 893 , and 742 cm^{-1}) were identified from those of the support, nearly aligning with those of PET (Tariq et al., 2020). Thus, the results confirm that the SFW was composed of PP (as a filter) and PET (as a support).

To clarify the compositional matrices of filtered particles and two separate components (filter and support) in the SFW, their thermolytic behaviours were analysed using a TGA instrument. A TGA run under inert (N_2) conditions was carried out using the designated temperatures: $35\text{ }^{\circ}\text{C}$ to $900\text{ }^{\circ}\text{C}$ at a steady rate with $10\text{ }^{\circ}\text{C min}^{-1}$. The TGA profiles of filtered particles, filter layer, support layer, and SFW are presented in Fig. 2(a). For comparative analysis, the TGA results for PP and PET are also shown in Fig. 2(a).

In the TGA test of filtered particles from the SFW, their mass reduction was developed in the temperatures of 227 – $336\text{ }^{\circ}\text{C}$, exhibiting the peak degradation rate curve at $267\text{ }^{\circ}\text{C}$. The residual mass of the filtered particles at $900\text{ }^{\circ}\text{C}$ was 47.2 wt%, likely attributed to the considerable mass of inorganics. This is supported from the TGA of filtered particles in air condition, showing a 37.9 wt% of final mass (Fig. S1). The mass of inorganics in the filtered particles was measured using an ICP analyser, revealing 2.61 wt% Ca, 1.62 wt% Fe, 1.17 wt% of Na, 1.09 wt% of Al, 0.62 wt% of K, 0.50 wt% of Mg, and 0.13 wt% of Zn. However, cumulative inorganic mass was 7.74 wt%, which differs

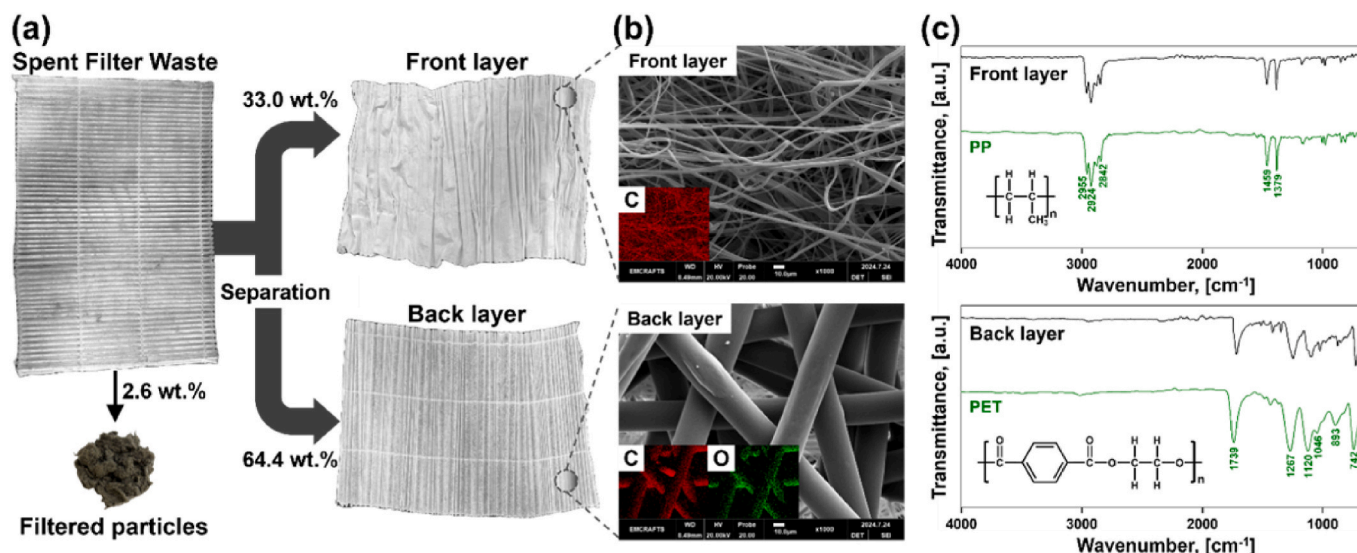


Fig. 1. (a) Filtered particles and front/back layers separated from SFW, (b) their surface morphology and elemental distribution (examined using a SEM/EDX unit), and (c) their FT-IR spectra (incorporated with those from polypropylene (PP) and polyethylene terephthalate (PET) as references).

significantly from the ash content (37.9 wt%) from Fig. S1. As such, this discrepancy is likely explicable due to the presence of Si-based substances, as their quantification is challenging through the ICP test. This inference is supported well by the fact that quartz wool, composed of SiO₂, is commonly used in the laboratory environment. Notably, the thermolytic trends for filtered particles was much different with the results from the SFW, noting that it was thermally decomposed in the temperature window of 370–490 °C. The inconsistency is likely attributed to the minor presence of filtered particles (2.6 wt%) within the SFW. The mass reduction of the filtered particles in SFW cannot be clearly observed due to their negligible mass reduction, equivalent to less than 0.6 wt%. These observations signify that filtered particles did not significantly influence the thermolysis patterns of SFW.

Meanwhile, a total mass reduction of the filter in the SFW was observed between 370 °C and 500 °C, with the highest thermolytic rate occurring at 456 °C. This indicated that the thermal degradation of the filter was complete by 500 °C. Similar thermolytic patterns were observed in the TGA results for PP. The consistency of the temperature associated with the fastest degradation rate confirmed that the filter was composed of PP. For the support layer, considerable mass reduction was observed between 400 °C and 483 °C, with a peak thermolytic rate at 436 °C. These thermolytic behaviours closely resembled those of PET, confirming that PET was the primary polymeric component of the support material. The thermal decomposition of the support was notably slowed at temperatures ≥ 483 °C, likely due to dehydrogenation and subsequent carbonisation of the remaining residues after the initial volatilisation (Fan et al., 2021). Given that the SFW consisted of 33.9 wt % of filter and 66.1 wt% of support, its overall thermal decomposition was expected to reflect the thermolytic patterns of both components. Despite the presence of these two distinct components, the TGA results showed substantial mass reduction in the temperature range 400–490 °C, with a characteristic thermolytic rate at 438 °C. This overlap in the thermal degradation temperature ranges of PP and PET likely contributed to this observation. Considering that the thermal decomposition of PP was complete below 500 °C, the slow mass reduction observed at temperatures ≥ 490 °C was attributed to the carbonisation of PET.

To further confirm the composition of the SFW matrices, a Py-GC/MS test was conducted. This system integrated a pyrolysis device with a GC/MS instrument, allowing for the identification of chemical constituents volatilised during thermal decomposition. The distribution of thermolytic products varied depending on the polymer type owing to inherent

differences in the chemical structure (Mondal et al., 2024). Both components, along with the SFW, were pyrolysed at a temperature of 600 °C, and the resulting VOCs were analysed in real-time using the GC/MS unit. The GC/MS chromatograms obtained from the filter, support, and SFW are provided in Fig. 2(b), highlighting the key constituent chemicals identified using the GC/MS analysis.

The volatile chemicals released from Py-GC/MS run of the filter particles showed a significant complexity, closely related to the VOC formations arising from experiments conducted in the laboratory. However, their identification in the pyrolytic product distribution from the SFW was not confirmed due to trace amounts of filtered particles. The Py-GC/MS test of the filter later primarily produced the branched hydrocarbons with a broad range of carbon numbers (C₃–C₆₁). The dominant presence of propylene and 2,4-dimethyl-1-heptene (C₉) indicated that the filter was composed of PP (Shin et al., 2011). The thermal degradation of the support layer produced chemical constituents with three characteristic functional linkages: acids (terephthalic acid and benzoic acid), esters (vinyl benzoate and divinyl terephthalate), and aromatics (benzene). These pyrolytic product distributions are consistent with PET results (Shin et al., 2011). The Py-GC/MS test outcomes for the SFW exhibited a complicated product distribution, involving primary chemicals arising from the thermal decomposition of PP and PET. Thus, it was confirmed that the two polymers in the SFW underwent simultaneous thermal decomposition because of their similar thermal stabilities, as supported by the pyrolytic product distribution.

3.2. Thermolytic trends of SFW in CO₂ conditions

To scrutinize the CO₂ effect on the thermolytic patterns of the SFW, TGA was performed in a CO₂ environment. The temperature program for the TGA test ranged from 35 to 900 °C at a steady rate of 10 °C min⁻¹. For an effective assessment of the CO₂ impact, the mass change and thermolytic rate curves of the SFW obtained under N₂ and CO₂ conditions were obtained, as provided in Fig. 3(a).

The difference in the thermal decomposition patterns of SFW from both N₂ and CO₂ environments was negligible at temperatures ≤ 800 °C. The similarity in TGA profiles suggested that CO₂ had no remarkable effect on the thermal decomposition of SFW. However, at temperatures ≥ 800 °C, a slight decrease in the residual mass was observed in the presence of CO₂. This was likely due to interactions between CO₂ and carbon-based residues, given that the residue was related to the carbonisation of the PET-based fabric in SFW. The heterogeneous reaction

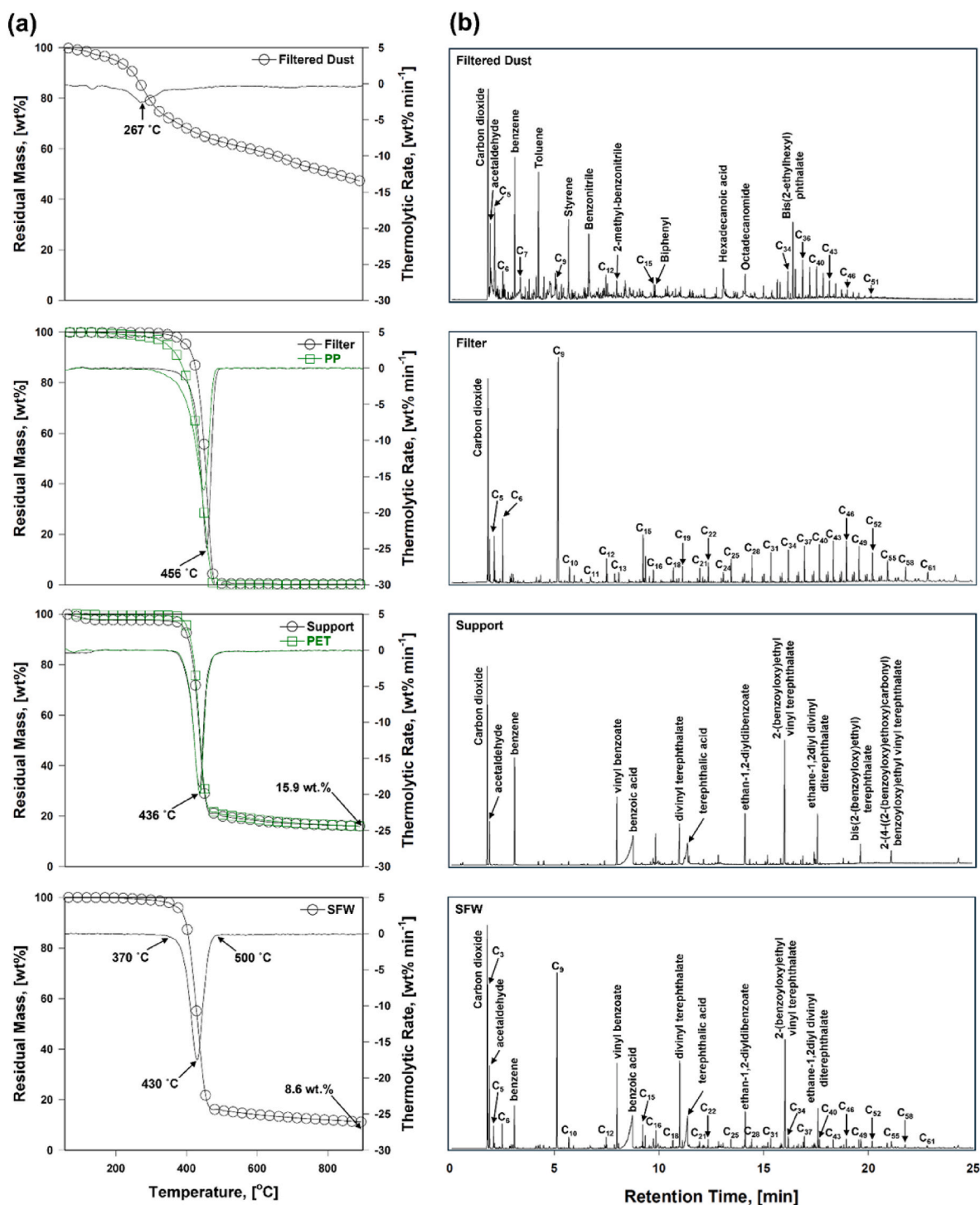


Fig. 2. (a) Mass changes of filtered particles, filter layer, support layer, and SFW; their thermolytic rates under N₂ conditions; and (b) their thermolytic product distribution identified using Py-GC/MS analysis.

could be explained using the Boudouard reaction, presented as $\text{CO}_{2(g)} + \text{C}_{(s)} \rightleftharpoons 2\text{CO}_{(g)}$ (Zhao et al., 2024). While the temperature for activating the Boudouard reaction is thermodynamically 700 °C (Arauzo et al., 2020), it appeared at 800 °C. This discrepancy indicated that the Boudouard reaction kinetics may be slow. To identify the degree of the Boudouard reaction, oxidation of SFW was implemented using the TGA test (Fig. 3(b)), leading to a negligible mass at 900 °C. This indicated that the Boudouard reaction was not fully activated at 900 °C while there was approximately 7.9 wt% of unreacted carbon remaining. Thus, the significant difference in the final residual mass between the CO₂ and air

atmospheres confirmed that the heterogeneous reaction kinetics associated with CO₂ were slow. While the occurrence of the heterogeneous reaction driven by CO₂ is presented in Fig. 3(a), there was no evidence of a CO₂ interaction at ≤800 °C. Considering that the TGA outcomes only address mass changes at functioning reaction temperatures, it is necessary to further investigate any homogeneous interactions of CO₂ with the VOCs evolved from the pyrolysis of SFW.

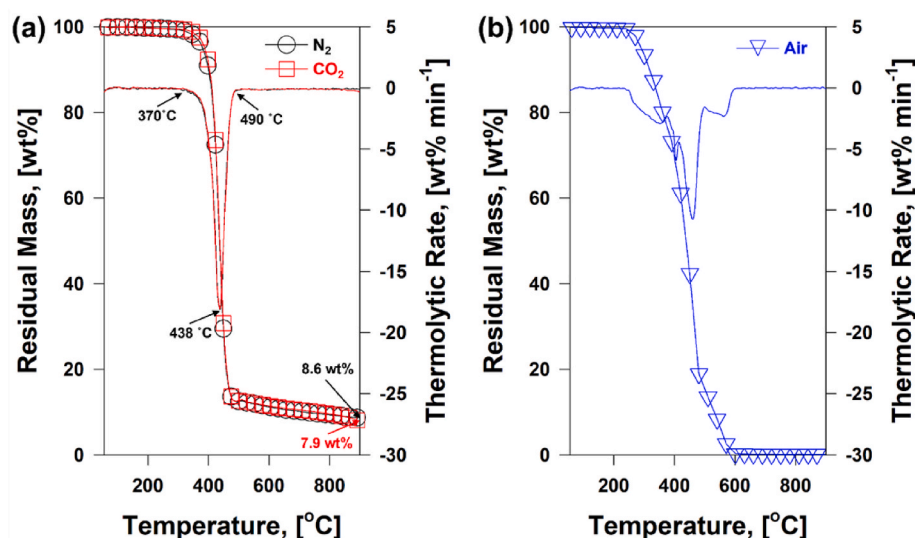


Fig. 3. Mass change and thermolytic rate curves of SFW under three different atmospheres: (a) N₂ and CO₂, and (b) air.

3.3. Impact of CO₂ on the volatile compounds from SFW

To investigate the effect of CO₂ on VOCs liberated from SFW, non-condensable (gaseous) and condensable (liquid) products were analysed qualitatively and quantitatively. Notably, the phase of the VOCs is a determinant of their inherent carbon length. The laboratory-scale pyrolysis for SFW was conducted in the designated temperatures: 100–700 °C at a steady rate with 10 °C min⁻¹. End temperature, equivalent to 700 °C, was chosen to block any effectiveness arising from the Boudouard reaction. The concentrations of the key pyrolytic gases, particularly H₂, CO, and CH₄, under N₂ and CO₂ atmospheres were measured. Their evolution patterns are plotted in Fig. 4. Cumulation of gaseous product concentrations obtained from SFW pyrolysis is also presented in Fig. 4.

In a N₂ environment, the H₂ evolution increased as the reaction temperature increases in the range of 490–670 °C. The temperature window for H₂ production corresponded well with the TGA results (Fig. 2(a)) that show that the retarded thermolytic rate at ≥490 °C is associated with dehydrogenation. The dehydrogenation's extent was in proportion to the test temperature (Geng et al., 2023). However, H₂

concentration was maximised at 670 °C and decreased at 700 °C. This was because of the depletion of the hydrogen source in the batch-type pyrolysis setup. A notable increase in the CO production was initiated at 310 °C and reached the highest concentration at 400 °C. Given that the PET-based fabric in SFW contained only oxygen-containing linkages in its polymeric backbone, the CO evolution was likely attributed to the deoxygenation of the VOCs evolved from PET pyrolysis. The formation of CH₄ was initiated at ≥310 °C, with the heightened concentration at 430 °C. This was closely related to the thermal decomposition of the PET- and PP-based polymeric backbones in the SFW.

When incorporating CO₂, overall evolution trends of H₂, CO, and CH₄ were considerably identical to those produced in an inert (N₂) environment. However, the concentration profile for CO in a CO₂ environment at temperatures ≥600 °C differed from that in a N₂ environment. A slight increase in CO formation was observed, with no change in the profiles of H₂ and CH₄. This observation could not be explained by the additional inputs of carbon and/or oxygen sources. Thus, CO₂ might have homogeneously interacted with VOCs evolved from the SFW. Partial oxidation of VOCs induced by CO₂ facilitated CO production. The increase in CO formation when injecting CO₂ was also

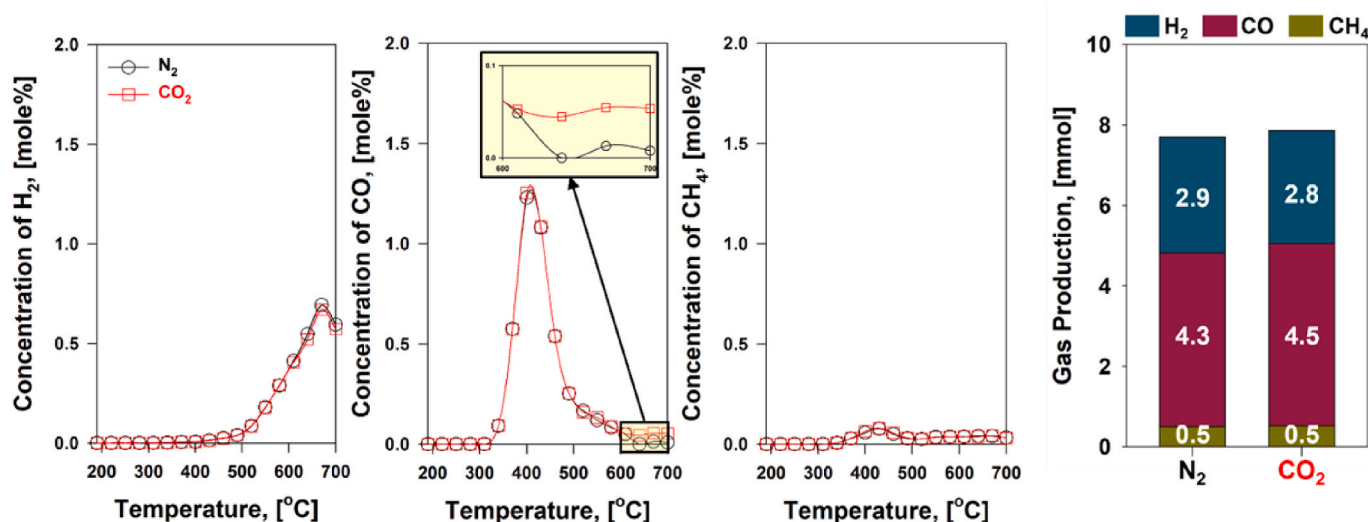


Fig. 4. Evolution profiles and cumulative production of key pyrogenic gases, H₂, CO, and CH₄, generated from SFW's conventional pyrolysis under 2 atm: N₂ and CO₂.

confirmed by the cumulative gas production (Fig. 4). Given the influence of CO₂ on VOCs, it modified compositional matrices by converting liquid products into gaseous products.

To confirm this, a qualitative investigation of the pyrolytic oil generated from the SFW pyrolysis in both atmospheres was implemented by a GC/MS test. The resulting chromatogram from N₂ environment is presented in Fig. 5(a). Of note, the chromatogram obtained under CO₂ conditions was not visualized because of its close resemblance to that obtained under N₂ conditions. Qualification information on the chemical distribution, having characteristic peak intensities, is provided in Fig. 5(a). The qualified chemicals were grouped into branched hydrocarbons, acids, esters, and aromatics, on the basis of their functional linkages. Cumulation in peak areas of the grouped chemicals are shown in Fig. 5(b).

The chemical composition of pyrolytic oil was closely related to the two polymeric components in the HEPA filter waste, primarily composed of PP and PET. The thermolysis of the PP-based layer predominantly produced branched hydrocarbons owing to the presence of methyl groups attached to the polymeric chain. The cumulative peak areas for branched hydrocarbons accounted for 56.6 % of the identified constituent chemicals. Branched hydrocarbons can potentially be used as transportation fuels. However, the direct utilisation of pyrolytic oil derived from SFW is challenging because of its wide hydrocarbon distribution (C₉–C₄₃). To acquire hydrocarbon fractions compatible with commercial fuels (gasoline: C₅–C₁₃ and diesel: C₈–C₁₈), a refinery process for pyrolytic oil is required. This is considered a critical factor in lowering economic feasibility (Yáñez et al., 2020). Additionally, pyrolytic oil derived from SFW contains other chemical groups such as acids, esters, and aromatics. Their formation was attributed to the thermal decomposition of the PET-based layer, accounting for 43.4 % of the peak area accumulation of the identified ones. Particularly, the abundance of aromatic chemicals retards their diffusion flame rate with an oxidant (air), adversely affecting their practical application as a liquid fuel for commercial engines. Thus, converting liquid pyrolytic products into gaseous products is desirable for sustainable valorisation of SFW.

In a CO₂ environment, the peak intensities of the constituent chemicals qualified in the pyrolytic oil were notably similar to those observed in a N₂ environment (data not shown for better visualisation). The negligible difference in the compositional matrices from both atmospheres indicated that the homogeneous interaction between CO₂ and VOCs was not activated. Considering the major volatilisation of SFW at 370–490 °C (Fig. 2(a)), the gaseous-phase interaction between CO₂ and the VOCs derived from SFW did not supply sufficient energy for activation. Therefore, to explore the functional reactivity of CO₂, strategic

measures are required to expedite the gaseous-phase interactions governed by CO₂.

3.4. Improved impact of CO₂ through catalytic pyrolysis

To expedite the gaseous-phase interaction of CO₂ with the VOCs from SFW, modification of conventional test setup was performed by additionally installing an isothermal heat block of 700 °C. The temperature, 700 °C, was chosen to avoid any appearance of the Boudouard reaction, as this reaction in SFW was activated at 800 °C. At the centre of the added heat block, two catalysts (Al₂O₃ or 5 wt% Ni/Al₂O₃) were located. The catalytic pyrolysis setup was intended to further crack the VOCs evolved from the conventional pyrolyser, while also providing favourable conditions for a homogenous reaction with CO₂. SFW's catalytic pyrolysis over two different catalysts was carried out under both atmospheres. The resulting major gaseous products were then quantified. H₂, CO, and CH₄ concentration trends from SFW pyrolysis in the presence of Al₂O₃ and 5 wt% Ni/Al₂O₃ are given in Fig. 6(a) and (b). Moreover, the accumulated gas generation is provided in Fig. 6(a) and (b), respectively.

In Fig. 6(a), overall gas concentration levels under N₂ conditions were enhanced at <600 °C, compared to those under the conventional pyrolysis setup. This resulted from the experimental modification by supporting additional heat (700 °C) and the Al₂O₃ catalyst. The concentrations for H₂ and CO were heightened at 430 °C, corresponding to the temperature at which major volatilisation occurred. This observation was closely linked to the expedited dehydrogenation and deoxygenation of VOCs. Similarly, the evolution patterns for CH₄ reached the highest concentration, likely attributable due to the enhanced thermal decomposition of the VOCs released from the SFW pyrolysis.

When adopting CO₂ as the reactive medium, the concentration patterns for H₂ and CO changed over the entire temperature range. When the H₂ production was reduced in a CO₂ condition, the CO formation is enhanced. The concurrent change in the presence of CO₂ suggests a reverse water-gas shift reaction (rWGS rxn.) (He et al., 2023). The rWGS rxn. is driven by an equivalent consumption of H₂ for CO production. However, the added formation of CO (4.48 mmol) exceeded the H₂ suppression (4.0 mmol). The surplus formation of CO may have resulted from other reaction routes associated with CO₂, suggesting a homogeneous reaction with VOCs from SFW.

To further accelerate the impact of CO₂ on the SFW-derived VOCs, 5 wt% Ni/Al₂O₃ catalyst was introduced. Fig. 6(b) presents the quantifying profiles of H₂, CO, and CH₄. The Ni catalyst was synthesized using the eggshell-type preparation method, positioning Ni at the surface layer

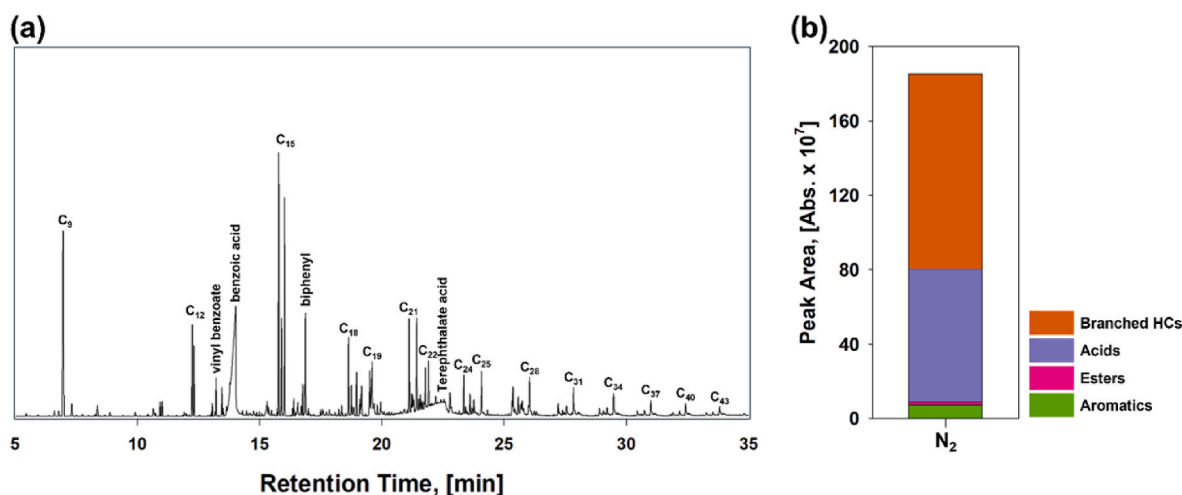


Fig. 5. (a) Chromatogram of pyrolytic oil from SFW pyrolysis under N₂ conditions, and (b) cumulative peak areas of chemicals grouped in the identified constituent chemicals.

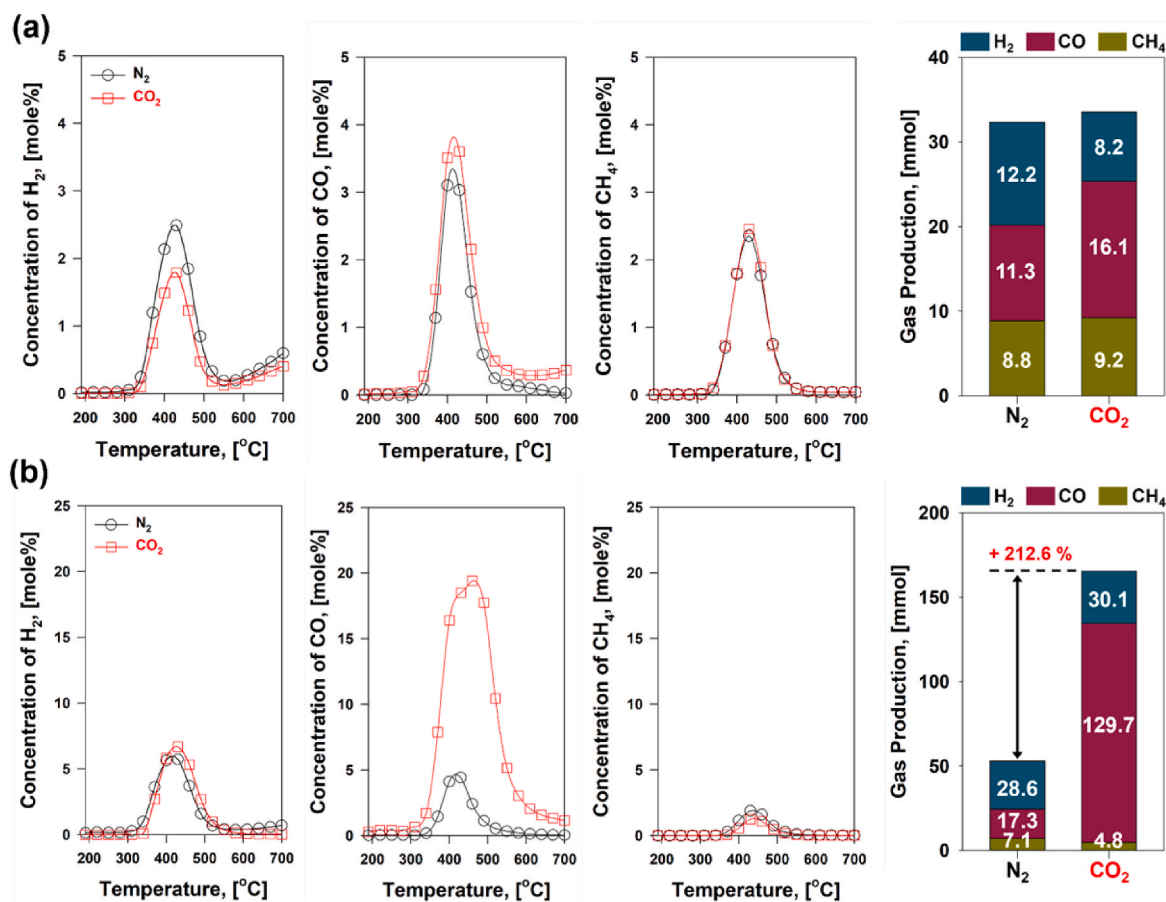


Fig. 6. Evolution profiles and cumulative production of key pyrogenic gases, H₂, CO, and CH₄, generated from SFW's catalytic pyrolysis over Al₂O₃ and 5 wt%/Al₂O₃ in N₂ and CO₂ atmospheres.

of catalyst. The distinctive structural formation showed technical benefits in shortening a diffusion path of the reactants, accelerating reaction kinetics. Indeed, the introduction of Ni catalyst under N₂ conditions further increased the H₂ and CO concentration levels, showing 2.3 times and 1.4 times, respectively, in reference to the results over Al₂O₃ catalyst. This is likely attributable to the chemical bond scission ability of Ni/Al₂O₃ (Si et al., 2023). However, the effectiveness of the Ni catalyst in the presence of CO₂ alters the H₂ and CO evolution patterns. Unlike the suppressed H₂ induced by the rWGS rxn. over Al₂O₃, the synergistic effect of Ni catalyst with CO₂ further expedited dehydrogenation. Simultaneously, the notable improvement in CO led to maximum concentrations (19.4 mol%), representing a 5.4-fold increase. The cumulative gas production from CO₂-assisted catalytic pyrolysis over Ni/Al₂O₃ was 164.6 mmol, increased by 212.6 % in reference to N₂ conditions. The contribution of CO₂ for syngas production was greater in CO production (129.7 mmol) than hydrogen production (30.1 mmol), which increases 5.2% for H₂ and 649.7% for CO as compared to the results over Al₂O₃ catalyst. These findings indicate that the functional reactivity of CO₂ for interactions with VOCs can be promoted over Ni/Al₂O₃, resulting in a significant CO generation.

The CO₂'s influence on syngas enhancement during catalytic pyrolysis can affect the compositional matrices of pyrolytic oil derived from SFW. To explore this, GC/MS analysis of the pyrolytic oil was performed. Accumulated peak areas for the grouped chemicals in the pyrolytic oil produced from catalytic pyrolysis over Al₂O₃ and Ni/Al₂O₃ under both atmospheres are shown in Fig. 7. The GC/MS chromatograms indicating peak intensity of constituent chemicals are visualized in Fig. S2. Detailed qualification results are summarised in Table S1. Qualified chemicals can be grouped into acids, aromatics, and ketones.

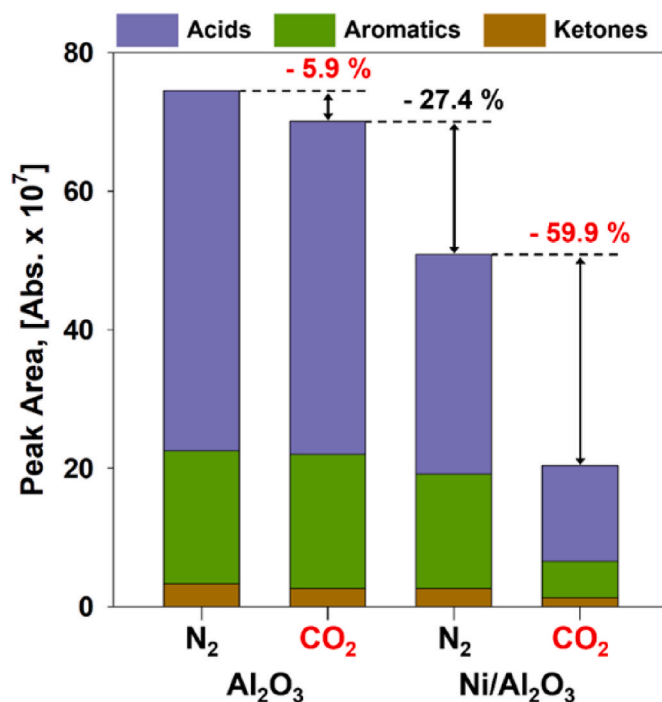


Fig. 7. Accumulated peak areas for the grouped chemicals in SFW-derived pyrolytic oil obtained from SFW's catalytic pyrolysis over Al₂O₃ and Ni/Al₂O₃ under two environments: N₂ and CO₂.

The disappearance of branched HCs, identified by conventional pyrolysis, could be due to the enhanced thermal cracking of the VOCs from the catalytic test setup and their subsequent recombination for aromatic formation. The incorporation of CO₂ during the catalytic pyrolysis resulted in a reduction of the resulting chemical groups. For the Al₂O₃ catalyst, a 5.9% reduction in peak area was observed under CO₂ conditions as compared to that N₂ conditions. When Ni metal was introduced into Al₂O₃, the effectiveness of CO₂ for the compositional modification improved. Cumulative peak areas of identified chemical groups decreased by 59.9% over Ni/Al₂O₃. These experimental findings support the assumption that the enhanced reactive functionality of CO₂ provides favourable conditions for its interaction with VOCs, resulting in CO enhancement.

For an in-depth exploration, the use of various CO₂ concentrations as the reactive medium was examined for the catalytic pyrolysis of SFW. Given that the catalytic reaction is driven by the collision frequency, adjustment of the CO₂ concentration helps optimise the CO₂'s functionality for syngas enhancement. Accumulated productions of H₂, CO, and CH₄ were obtained under 0-, 20-, 50-, and 80-vol.% CO₂ conditions (on the balance with N₂), provided in Fig. 8(a). When purging with a 20 vol% CO₂, a significant increase in the syngas production was identified, equivalent to 153.4%. Subsequently, the total gas production was proportional to the CO₂ concentration. This suggests that an increase in CO₂ concentration provides opportunities for its gaseous phase to interact with VOCs, thereby converting them into CO. The functional reactivity of CO₂ is evident from the product balance by mass presented in Fig. 8 (b). The GC/MS chromatograms of pyrolytic oils from catalytic pyrolysis under diverse CO₂ concentrations are visualized in Fig. S3. Detailed qualification results on the chemical constituents are summarised in Table S2. The transformation of liquid products into gaseous products intensified as a function of CO₂ concentration. All the experimental findings in this study prove that the mechanical features of CO₂ are governed by its homogeneous reaction with SFW-derived VOCs and its subsequent conversion into CO. Although catalytic pyrolysis of SFW under diverse CO₂ concentrations facilitated identifying the effectiveness of CO₂, it is necessary to additionally investigate the impact of other process parameters, such as reaction temperature, Ni loading of catalyst, catalyst-to-feedstock ratio, and others. Therefore, to further optimise the functional reactivity of CO₂, the following studies should be conducted in the near future.

The use of Ni catalyst in the catalytic pyrolysis of SFW from N₂ and CO₂ conditions enhanced the syngas production. Given the notable activation of Ni for the dehydrogenation of VOCs, the Ni catalyst can be readily deactivated due to the coke formation, thereby reducing catalytic capability. To explore the degree of coke formation for two used catalysts from N₂ and CO₂ conditions, their temperature-programmed

oxidation (TPO) tests were assessed. The TPO test is an effective method to evaluate the catalyst stability. The TPO results from pristine catalyst, and two used catalysts from N₂ and CO₂ were provided in Fig. S4. The amount of coke (11.9 wt%) from the N₂ condition was much higher than that (5.2 wt%) from the CO₂ condition. This finding supports that the use of CO₂ in the catalytic pyrolysis offers opportunities to block the coke formation while extending the lifespan of Ni catalyst.

3.5. Potential to reduce CO₂ emissions from CO₂-assisted catalytic pyrolysis

The reactive functionality of CO₂, which enables its gaseous-phase homogeneous interaction with VOCs and their transformation into CO, was demonstrated by modifying the test setup and adjusting the CO₂ concentration as a reactive medium. CO₂-assisted catalytic pyrolysis over Ni/Al₂O₃ exhibited superior performance for CO production, indicating that the largest amount of CO₂ can be consumed. To assess the sustainability of the proposed pyrolysis test setup, its potential to reduce CO₂ emissions was compared with a traditional disposal method (incineration) and catalytic pyrolysis under N₂ conditions. For easier estimation of CO₂ emissions for the three different processes, it was assumed that 1 million tons (Mt) of SFW was processed. The CO₂ emissions estimates for each treatment process are shown in Fig. 9.

To calculate the CO₂ emissions from incineration, the elemental analysis results of the SFW were used. Considering that the SFW contains a 71.03 wt% of carbon, it was converted as CO₂-equivalent, determining 23.4 Mt of CO₂ emissions when assuming the complete oxidation. During the catalytic pyrolysis in a N₂ environment, the cumulative production of CO₂ was 6.1 mmol, which was primarily attributed to the thermal decomposition of the PET-based layer in the SFW. Thus, the considerable generation of CO₂ emissions (equivalent to 0.8 Mt) was estimated when conducting catalytic pyrolysis without the CO₂'s input as a reactive feedstock gas. However, employing CO₂ in catalytic pyrolysis led to its consumption by interacting with SFW-derived VOCs, resulting in CO production. To estimate the CO₂ consumption, the homogeneous reaction pathway driven by CO₂ was assumed as 'initial VOCs (C_aH_bO_c) + CO₂ → reacted VOCs (C_{a-1}H_bO_c) + 2CO'. Surplus CO from catalytic pyrolysis under CO₂ conditions was 112.4 mmol in reference to N₂ conditions. This indicates that the homogeneously interacting CO₂ was estimated as 56.2 mmol, signifying the suppression of 6.6 Mt atmospheric CO₂. Based on these calculations, a comparison of the CO₂ emissions confirmed that the proposed pyrolysis setup (CO₂-assisted catalytic pyrolysis) is the most sustainable practice for SFW valorisation. Therefore, this study suggests that CO₂ utilisation from the catalytic process transforms SFW into a combustible fuel, particularly CO, while contributing to global warming mitigation.

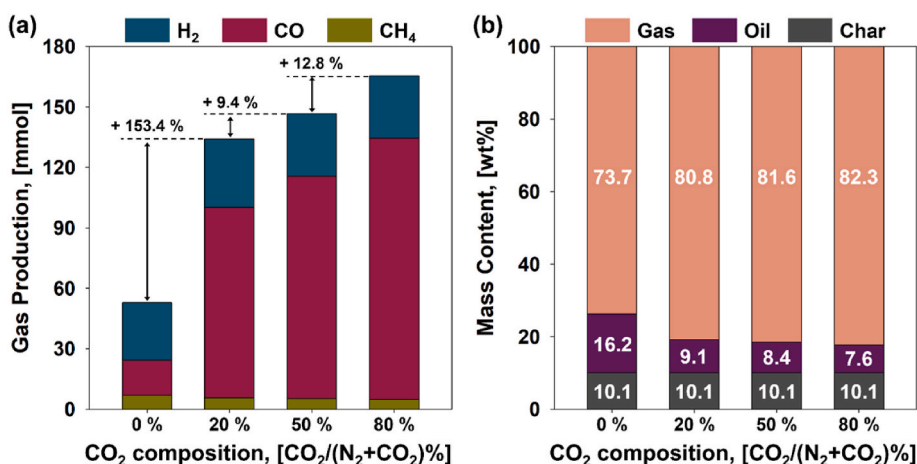


Fig. 8. (a) Cumulative gas production, and (b) the product balance resulted from SFW catalytic pyrolysis in varied CO₂ concentrations.

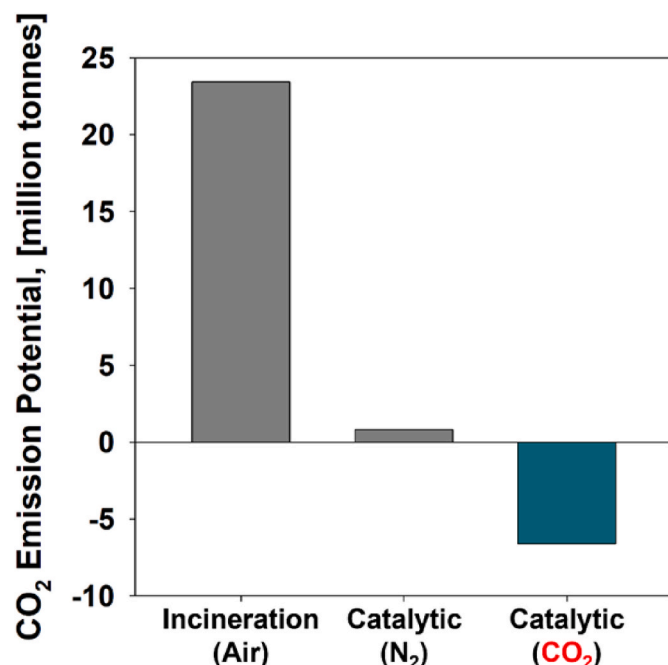


Fig. 9. Potential of CO₂ emissions from three different practices (incineration and catalytic pyrolysis over Ni/Al₂O₃ in both N₂ and CO₂ atmospheres) for the disposal of SFW.

4. Conclusions

This study explored the development of a thermochemical platform for SFW valorisation, especially for the SFW valorisation. To impart the improved sustainability to pyrolysis process, the CO₂'s employment was made as a reactive feedstock gas. Before the pyrolysis test, several characterisations of the SFW were conducted, including SEM, FT-IR, TGA, and Py- GC /MS analyses, confirming that the SFW consisted of a PP-based filter layer and a PET-based support layer. When adopting CO₂ to the pyrolysis process, it interacts with the SFW-derived volatiles, converting them into CO. However, the impact of CO₂ on the liquid and gaseous products was not notable, with a slight increase in CO under N₂ conditions. However, catalytic pyrolysis significantly enhanced syngas production, especially that of CO, with the expedited homogeneous interaction of CO₂ over the Ni/Al₂O₃ catalyst. The transformation of SFW into CO increased with increasing CO₂ concentration, which determined the reactive functionality of CO₂. To evaluate the sustainability of the proposed pyrolysis system, CO₂ emissions were compared to the results from traditional waste-to-energy methods, such as incineration and catalytic pyrolysis. Therefore, the experimental results demonstrate that the proposed pyrolysis system in this study offers opportunities for producing combustible fuels, particularly CO, and simultaneously has potential for global warming alleviation.

CRediT authorship contribution statement

Taewoo Lee: Writing – original draft, Validation, Formal analysis, Data curation. **Jung-Hun Kim:** Writing – original draft, Visualization, Methodology, Investigation. **Jaewon Lee:** Writing – review & editing, Validation, Data curation. **Eilhann E. Kwon:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.chemosphere.2024.143863>.

Data availability

Data will be made available on request.

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